HaDy_MZB

Habitat Dynamics MacroZooBenthos

A Python toolbox for patch-scale habitat dynamics analysis

User manual

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Aude Lecrivain, Christine Weber, Nico Bätz

Eawag, Swiss Federal Institute of Aquatic Science and Technology, Surface Waters – Research and Management, Kastanienbaum, Switzerland

GitHub repository:

https://github.com/HaDy-toolbox/HaDy_MZB

DOI: XXXX

Impressum

Authors of the toolbox and user manual

Aude Lecrivain, Christine Weber and Nico Bätz

Scientific background

This toolbox implements the framework presented in: Lecrivain A, De Cesare G, Weber C, & Bätz N (in Review). Macroinvertebrate habitat dynamics under frequent hydropower-induced discharge fluctuations: patch-scale metrics to quantify effects of hydropeaking and morphological complexity. XXXX

Technical support

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Toolbox testing

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XXX

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Contents

1. Toolbox presentation

1.1 Purpose of the HaDy_MZB toolbox

Rivers are dynamic and habitat conditions change over time as discharge varies and interacts with channel morphology. These changes are especially relevant at the patch scale, where organisms experience hydraulic stress, habitat persistence, dewatering, and shifts between suitable and unsuitable habitat conditions.

HaDy_MZB is a free and open source Python-based HaDy-toolbox for quantifying flow-driven habitat dynamics at the patch scale, with a focus on benthic macroinvertebrates—organisms particularly vulnerable due to their limited mobility, complex life cycles, and sensitivity to hydraulic stress. However, the toolbox can also be adapted to other organisms such as fish eggs, larvae, or vegetation, which may likewise be affected by limited mobility or exposure to changing habitat conditions.

HaDy_MZB links flow time-series with precomputed hydraulic or habitat information for selected discharges to reconstruct habitat time-series. Based on these habitat time-series, the toolbox calculates spatially explicit metrics that describe habitat availability, habitat stability, and exposure to hydraulic stress (see Table 1). A patch represents a fixed spatial unit corresponding to the ecological perception of macroinvertebrates, typically defined by the computational mesh used in high-resolution hydrodynamic models.

The toolbox builds on the patch-scale approach developed by Bätz and applied in Lecrivain. In particular, it puts into practice the logic of deriving habitat time-series from flow scenarios and habitat distributions. Hydropeaking is a key application because frequent sub-daily flow fluctuations strongly increase habitat dynamics, but the toolbox can also be used more generally to analyze habitat responses to variable flow regimes.

Table 1 Metrics derived by the HaDy_MZB toolbox

Metric	Name	What it measures	Ecological meaning
M1	Habitat probability	Time-integrated availability of habitat conditions	Indicates habitat availability and dominance
M2	Habitat shifts	Frequency of shifts between habitats	Reflects habitat stability and persistency
M3	Drift risk	Exposure to hydraulically stressful conditions (e.g. high current velocity)	Indicates risk of drift
M4	Desiccation risk	Exposure to low water depth / drying conditions	Indicates risk of stranding/dessciation

Who HaDy_MZB is for:

HaDy_MZB is designed for **researchers** and **practitioners** who want to assess flow-driven habitat dynamics and ecological stressors in a spatially explicit way. It supports applications such as hydropeaking impact assessment, comparison of hydrological scenarios, evaluation of morphological complexity, and reach-scale mitigation planning.

The toolbox does not replace hydrodynamic modelling or ecological expert judgement. Instead, it provides a reproducible workflow for translating existing hydraulic or habitat data and flow time-series into interpretable habitat-dynamics metrics.

1.2 Overview of the Workflow

The execution of the toolbox involves 3 steps (Figure 1):

- **Pre-processing:** required input data (flow time-series, hydraulic or habitat data) are collected and formatted to prepare the analysis (see section 3). Based on these inputs, different scenarios can be configured and compared.
- **Processing**: the toolbox links flow time-series with hydraulic or habitat data to generate habitat time-series and calculate selected metrics.
- **Post-processing:** the resulting outputs can be visualized, mapped and further analysed using GIS software such as QGIS (see section 5 , and 9 GQIS tips and tricks).

A project is defined for a given study area where hydrodynamic simulations have already been carried out by the user. From the simulations, spatially distributed data on water depth and current velocity must be provided by the user for a representative range of discharges (input shapefile). These data are then used to derive habitat types. Alternatively, the user can directly provide a spatially explicit habitat classification for a representative range of discharges. In addition, a flow time-series is required to describe the temporal evolution of discharge and define the simulation timestep.

HaDy_MZB does not run hydrodynamic simulations. Instead, it uses existing hydraulic or habitat data and user-defined ecological criteria to quantify habitat dynamics in a reproducible way. The respective roles of the user and the toolbox are summarized in Table 2.

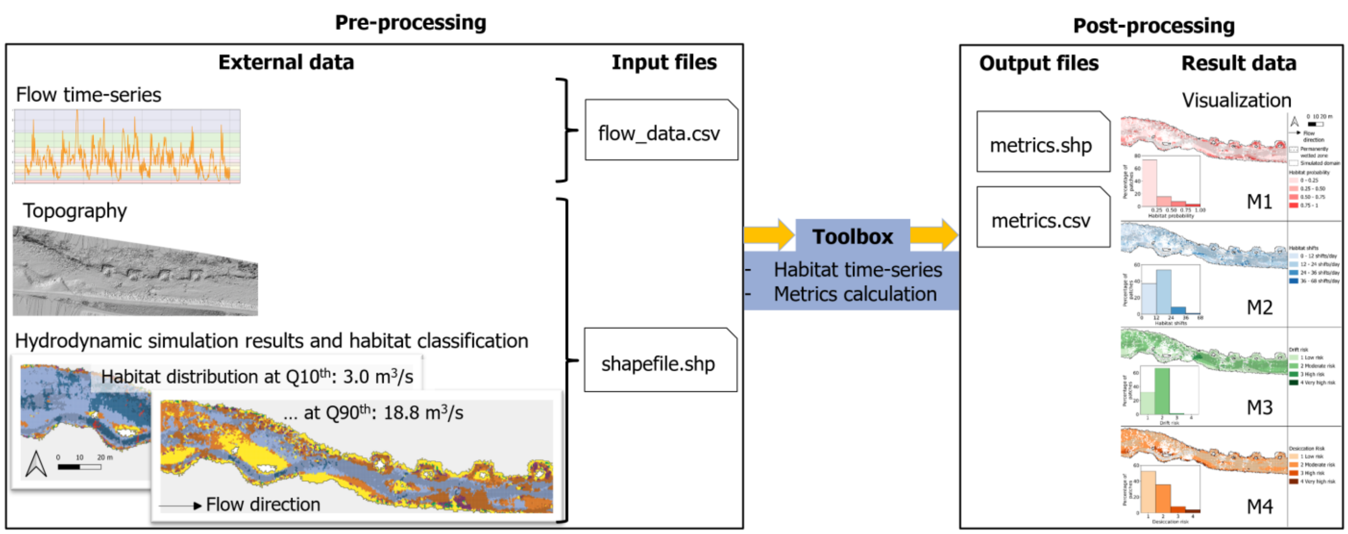


Figure 1 Overview of the execution procedure of the toolbox HaDy_MZB

Table 2 Respective roles of the user and the toolbox throughout the workflow

User responsibilities	
Pre-processing	Selects the flow time-series at the timestep of interest
	Selects associated discharges of interest
	Selects the zone of interest
	Runs the hydrodynamic simulation for the zone and representaive discharges
	Decides on a habitat classification method and relevant ecological thresholds
Post processing	Analyses the output on GIS software
	Adds additional plots or analysis if needed
Toolbox responsibilities	
Processing	Builds **habitat time-series** from flow time-series, water depth, current velocity and related habitat classification
	Calculates the selcted metrics for the habitat type of interest
	Exports results (i.e. metric values) in a shapefile and a csv document
	Provides graphical plots
	Provides additional functions for deeper analysis if needed

1.3 Example case study

This manual follows a **running example dataset** provided with the toolbox (Windows operating system). Blue boxes such as the following provide practical examples and shortcuts to quickly explore, reproduce, and adapt the workflow (see for instance 2.4 Example case study: from python repository to running the code and 5.4 Example case study: output presentation).

Running the example case as an initial analysis is a good way to become familiar with the toolbox.

Example case study: Quick Input–Output example (no setup required)

Example inputs (data > input, see Figure 2):

- Flow time-series (see Week_timestep_10_min.csv)
- Input shapefile with water depth and current velocity value (see Example_input_shapefile.shp → column v_discharge and d_discharge)
- Habitat definition (see section 4.3 Classify habitats of this user manual: classification entered in habitat_classification.py in the scripts)

Example outputs (data > output > Example_output, see Figure 2):

- Metrics M1–M4 (see metrics.csv)
- Maps (see metrics.shp and GIS_project)
- Plots (see folder Plots)
- Clustering (see folder Clustering)

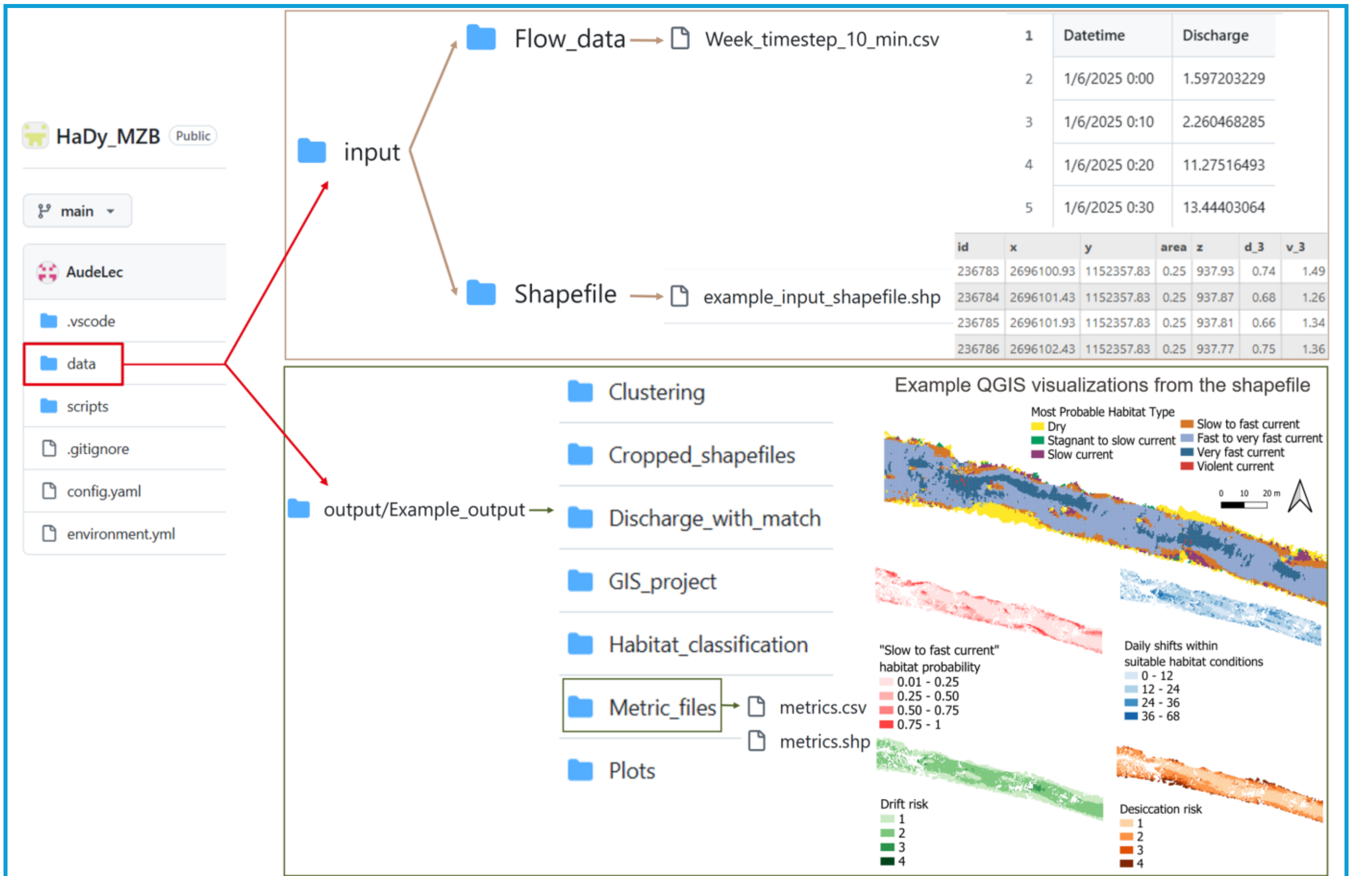


Figure 2 Quick input-output overview of the example case study GitHub

Click on the boxes Figure 3 to quickly get started with the toolbox, following the example case study blue boxes.

Quick start: key steps

2. Technical setup
3. Inputs + configuration
5. Output interpretation

Figure 3 Key steps to get started with the toolbox

2. Technical setup

This chapter explains how to install the toolbox and verify that everything works by reproducing the example dataset provided in the repository.

2.1 Prerequisites

If you do not have the following usual python core tools mentioned below and need support in installing them, check the **Appendix technical support file** provided in the repository.

Required tools:

- Python
- Git
- Anaconda/Miniconda
- Visual Studio Code (recommended), including the python extension
- Python extension for VS Code

This file also provides support for basic commands such as accessing a folder via the terminal.

2.2 GitHub repository setup

All steps are explained and illustrated for the example case study in the blue box 2.4 Example case study: from python repository to running the code.

Step 1: Create a working directory

Create a folder where the Git repository will be cloned.
This directory can be located anywhere on your system and named according to your preference.

Step 2: Clone the repository

Open a terminal and go to the created working directory: enter the command “cd” and the absolute path to your directory (**cd path/to/your/directory**).

Once in the working directory, paste the cloning command (Git needs to be installed otherwise this will not work, see 2.1 Prerequisites):
git clone https://github.com/HaDy-toolbox/HaDy_MZB.git

Step 3: Setup the virtual environment

In the terminal (for macOS and Linux), or anaconda prompt (Windows) go to **the cloned folder called HaDy_MZB** in your working directory, and paster following commands:
conda env create -f environment.yml, and then **conda activate Venv_HaDy_MZB**

Step 4: Open the project on VS code and select the python interpreter

Open the software VS code previously installed, open the cloned HyDy_MZB folder.
To select the interpreter: Ctrl+Shift+P (to get the command panel) → type “Python: Select Interpreter” → Choose Venv_HaDy_MZB.

2.3 Running the code

To run the main code, go the python file *main.py* and click the play button (see example box in 2.4 Example case study: from python repository to running the code).
The same steps can be done in the *plot.py* or *clustering.py* for further analysis (see section 4.6 Additional functions).

ATTENTION

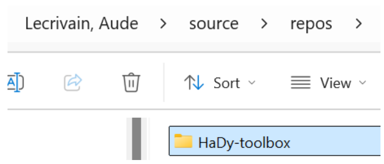
> If you tailor the analysis, make sure to have adapted **and saved** your local.yaml file (more in section 3.2) before running the codes, otherwise outputs will not match the entered parameters.

2.4 Example case study: from python repository to running the code

The following box explains and details the steps to follow to run the example case study and get familiar with the toolbox.

Example case study: From python repository setup to running the code

Step 1: Creation of a working directory



Step 2: Cloning the repository

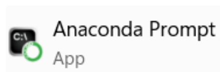
In the terminal, type in "cd" and paste the absolute path to your directory

```
PS C:\Users\lecrivau> cd C:\Users\lecrivau\source\repos\HaDy-toolbox
```

Then type in the cloning command

```
PS C:\Users\lecrivau\source\repos\HaDy-toolbox> git clone https://github.com/HaDy-toolbox/HaDy_MZB.git
```

Step 3: Setting up the virtual environment



On windows, use Anaconda prompt

Go to the cloned HaDy_MZB within your directory and type in the commands

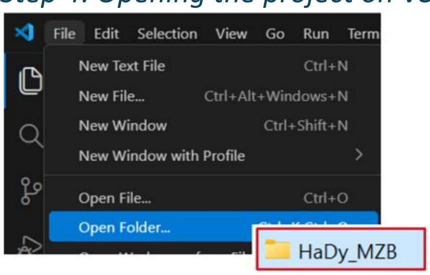
```
C:\Users\lecrivau\source\repos\HaDy-toolbox>cd C:\Users\lecrivau\source\repos\HaDy-toolbox\HaDy_MZB

C:\Users\lecrivau\source\repos\HaDy-toolbox\HaDy_MZB>conda env create -f environment.yml

rs\lecrivau\source\repos\HaDy-toolbox\HaDy_MZB>conda env create -f environment.yml
```

You now see that the environment is activated. `(Venv_HaDy_MZB) C:\Users\lecrivau>`

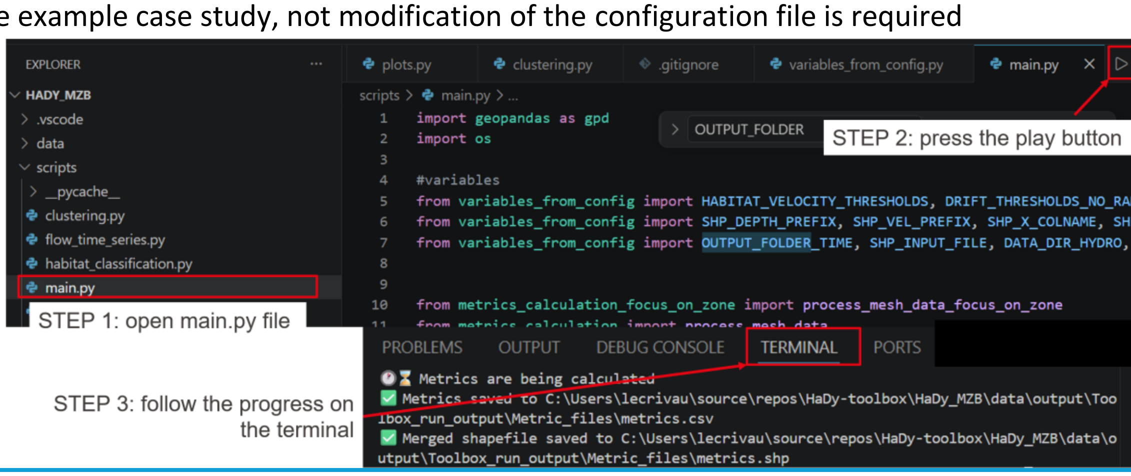
Step 4: Opening the project on VS code and selecting the python interpreter



To select the interpreter: Ctrl+Shift+P (to get the command panel) → type "Python: Select Interpreter" → Choose `Python 3.11.11 (Venv_HaDy_MZB) ~\.conda\envs\Venv_HaDy_MZB\python.exe`

Running the code

For the example case study, not modification of the configuration file is required



2.5 Later step: getting the latest version from the GitHub

Since the toolbox is in constant evolution, make sure to use the latest available. To update it, you need to "pull" the changes from the GitHub folder as done on Figure 4.

This can be done directly in the Visual Studio code terminal as follows: command **git pull origin main**

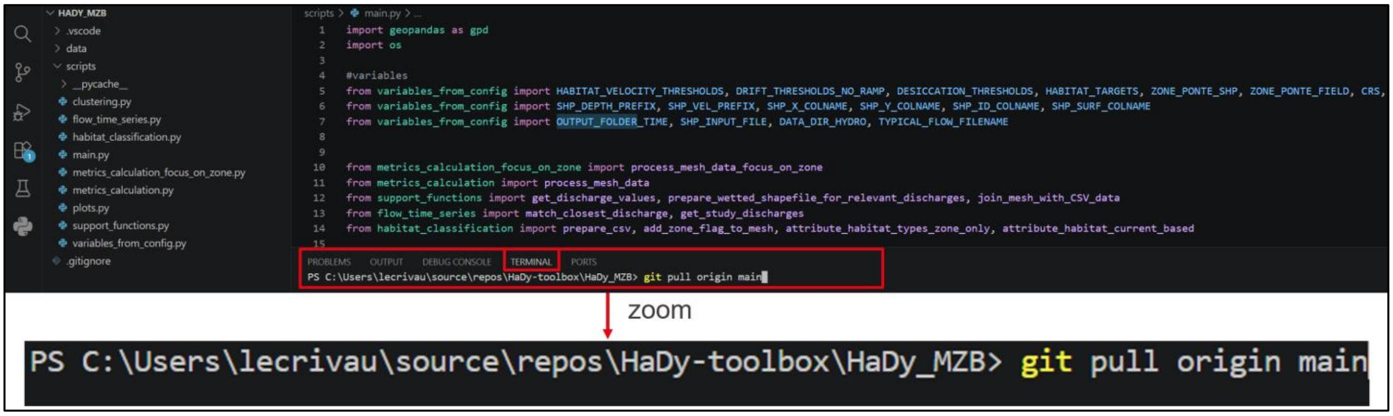


Figure 4 Example of how to pull the last version

ATTENTION

This command will overwrite your script folder and input data local folder with the latest content of the online GitHub repository, so **make sure to have saved ALL your changes in a safe space (e.g., a separate folder) before running this command**.

3. Inputs and configuration

3.1 Inputs files

Flow time-series input data

<u>Necessary minimum requirements</u>:

- csv file with variations of discharge (m^3/s) in time (e.g., Figure 5).
- The location is folder data > input > Flow_data.
- The column containing the discharges need to be called "Discharge" (exactly like that, including the capital letter). The format of the other columns is not important, and can be kept or deleted (e.g., Figure 5, right). The date column has been kept for user friendliness but is never read by the code.
- Discharge values must be formatted as numbers, with decimals separated by a dot "."
- The timestep of the discharge data entered in the configuration file must match this file's timestep (more in section 3.2). The toolbox needs to know the timestep separating two lines, for instance to calculate daily metrics. **It is the user's responsibility to provide an input file matching the relevant analysis timestep.**

ATTENTION

<u>Common issues:</u>

- Wrong location of the file.
- File as excel instead of csv format.
- Wrong "Discharge" column name.
- Coma instead of point as decimal separator.
- Missing values.
- Unmatching timesteps leading to false metric values.

Date (TU)	Valeur (en mÂ³/s)	Statut	Qualification	MÃ©thode	ContinuitÃ©
2025-06-24T00:00:00.000Z	14.4	12	20	8	0
2025-06-24T00:10:00.000Z	14.4	12	20	8	0
2025-06-24T00:20:00.000Z	14.4	12	20	8	0
2025-06-24T00:30:00.000Z	14.4	12	20	8	0

Date (TU)	Discharge
2025-06-24T00:00:00.000Z	14.4
2025-06-24T00:10:00.000Z	14.4
2025-06-24T00:20:00.000Z	14.4
2025-06-24T00:30:00.000Z	14.4

Figure 5 Example of data downloaded from the HydroPortail platform (on the left), and what is kept and renamed for the analysis (right). The Date column is kept here for user friendliness but is never read by the code. It is therefore not mandatory.

Shapefile input data

<u>Necessary minimum requirements:</u>

- Polygon shapefile (.shp and associated other file extensions) with water depth and current velocity data for discharges of interest (e.g., Figure 8). This data generally comes from hydrodynamic simulations
- The location is folder data > input > Shapefile
- The exact column names prefixes must be entered in the configuration file (more in section 3.2)
- The shapefile needs to contain the data presented in Table 3.

Table 3 Data that needs to be in the input polygon shapefile

Id	Id of the patch - One row per patch
x	Coordinates of the patch center
y	
Area m2	Associated patch area (useful to normalize metrics to the area if it varies a lot)
Water depth for each discharge **(in meters)**	Depth values for each discharge of interest. Decimals separated by "_" e.g., d_31_8 and d_6 for respectively discharges 31.8 and 6 m^3/s if the prefix is "d_"
Water current velocity for each discharge **(in meters/second)**	Current velocity values for each discharge of interest. Decimals separated by "_" e.g., v_31_8 and v_6 for respectively discharges 31.8 and 6 m^3/s if the prefix is "v_"

id	x	y	area	z	d_1	v_1	d_2	v_2	d_3	v_3	d_4_2	v_4_2	d_5_5	v_5_5	d_7_4	v_7_4	d_9	v_9	d_10_6	v_10_6	d_12	v_12
276139	2696154.44	1152347.31	0.25	937.3	0.33	0.62	0.43	0.88	0.53	1.18	0.58	1.29	0.65	1.43	0.74	1.57	0.8	1.65	0.87	1.71	0.92	1.76
276140	2696154.94	1152347.31	0.25	937.3	0.34	0.6	0.45	0.87	0.55	1.17	0.59	1.28	0.66	1.42	0.75	1.56	0.82	1.65	0.88	1.71	0.93	1.76
276141	2696155.44	1152347.31	0.25	937.28	0.35	0.57	0.46	0.82	0.56	1.11	0.61	1.21	0.68	1.36	0.77	1.5	0.84	1.59	0.9	1.65	0.95	1.7

Figure 6 Example of data in the polygon shapefile

ATTENTION

Common issues:

- Missing files: if only the .shp file in the data > input > Shapefile, the toolbox will not proceed. The other necessary formats are required (see files in data > input > Shapefile > Example_input_shapefile).
- Empty or n/a values: to be set to 0 for the toolbox to run an analysis.
- Data consistency: for a patch having no water for a certain discharge (depth = 0), then the current velocity value also needs to be zero.
- Missing data: each discharge of interest needs to have current velocity and water depth data (metrics need both to be calculated, so just current velocity or just water depth will not be sufficient).
- Wong formatting: the water depth and current velocity fields need to have the value of the discharge of interest in their title. Floats need to be separated by a "_" (e.g., d_31_8 and v_31_8 or d_6 and v_6). The water depth and current velocity columns prefixes (before the discharge value) need to be identical between each water depth column, and between each current velocity column, and match what is entered in the configuration file (more in section 3.2).
- Continuous spatial representation of the results: it is necessary for the shapefile to be a polygon shapefile and not a point shapefile.

3.2 Configuration

Necessary minimum requirements:

- To run an analysis that is different than the example case study, a *local.yaml* needs to be created in the repository. It is the user’s responsibility to create this file by copy pasting the *config.yaml* and making the corresponding modifications to tailor the analysis (see relevant video).
- The location of *local.yaml* is the root of the repository, where *config.yaml* is located

ATTENTION

Common issues:

- Wrong name: It is the user’s responsibility to create the *local.yaml* (spelled exactly that way) file to tailor the parameters.
- No *local.yaml* file: If no *local.yaml* file exists, the toolbox is running the *config.yaml* which is the example case study by default.
- Analysis not matching the entered parameters: It is important to **save the file** once changes are made to run the code with the correct parameters.
- Losing the previous analysis: Each new toolbox run stores the outputs in data > outputs > Toolbox_run_output, overwriting what is already in the folder. Save the analysis files you want to keep in another folder or rename them to keep them.

Table 4 describes the variables that need to be adapted by the user. It is split in different categories.

Table 4 Detail of the configuration variables

Input files

Name of the variable	Signification	Status
input_shp_filename	Name of the input shapefile located in the data > input > Shapefile folder	Mandatory
crs	CRS of the GIS project	Mandatory
input_flow_timeseries_filename	Name of the flow time-series input file located in the data >input > Flow_data folder	Mandatory
time_step_min	Timestep of the flow time-series in minutes (float, e.g., 10.0 for a 10-minute timestep). /!\ Must be the one of the flow time-series input file	Mandatory
shp_id_colname	Name of the id column identifying each patch	Mandatory
shp_x_colname	Name the x coordinate column of each patch in the input shapefile	Mandatory
shp_y_colname	Name the y coordinate column of each patch in the input shapefile	Mandatory
shp_area_colname	Name of the column containing the area of each patch (for potential area metrics normalization)	Mandatory

shp_depth_prefix	Depth prefix before the discharge values in the input shapefile (e.g., “d_30_2” → the prefix is “d_”)	Mandatory
shp_vel_prefix	Current velocity prefix before the discharge values in the input shapefile (e.g., “v_30_2” → the prefix is “v_”)	Mandatory

Habitat classification thresholds

Name of the variable	Signification	Status
depth_threshold	Minimum water depth (in meters) to consider a patch wet (float, e.g., 0.01 for 1 cm, meaning that patches will be considered wet if water depth is greater than 1 cm)	Mandatory
habitat_velocity_thresholds	Example of habitat classification thresholds based on current velocity and water depth (see section 4.3 Classify habitats)	Mandatory
number_of_habitats	Integer: the number of habitats that you have based on the habitat classification (e.g., 7 for the example classification: from 0 = dry to 6 = v ≥ class_5 m/s)	Mandatory

ATTENTION

<u>Common issues:</u>

- Count correctly your number of habitats: e.g., from 0 to 6 means you have 7 habitat types. If it is not correct, plots won’t work.
- If you change the habitat classification method, please adapt the function attribute_habitat_current_based in habitat_ classification.py accordingly

Metrics calculation

Name of the variable	Signification	Status
target_habitat	Target habitat type for which the metrics will be calculated. Metrics are only calculated for the patches that experience this habitat type at least once during the habitat time-series Unique integer from the habitat classification definition. If you want results for different target habitats, you need to run the analysis several times.	Mandatory
metrics_to_compute	Metrics selection: "true" to compute the metric, "false" otherwise	Mandatory
shift_all_daily	Daily shifts between all types of habitats	Mandatory
shift_targ_daily	Daily shifts between the suitable_hab_shifts habitat types	Mandatory
shift_dry_daily	Daily shifts to dry conditions (water depth < depth_threshold)	Mandatory

dry_max	Maximum consecutive time steps of dry conditions (in number of timesteps)	Mandatory
desiccation_risk	Associated desiccation risk value (see section 4.4 Metrics calculation)	Mandatory
drift_percentile	Drift percentile risk (see section 4.4 Metrics calculation)	Mandatory
drift_max	Associated maximum drift value (see section 4.4 Metrics calculation)	Mandatory
drift_durations	Number of timesteps spent is each drift category	Mandatory
suitable_hab_shifts	List of habitat types considered “suitable” in between which shifts will be calculated --> shift_targ_daily	Mandatory
start_at_first_occurrence	"true" to start the desiccation metric calculation after the first occurrence of the habitat type in the habitat time-series. "false" to start the desiccation metric calculation at the beginning of the time-series	Mandatory
desiccation_thresholds	Example of desiccation thresholds based on consecutive durations spent outside of water (in hours)	Mandatory
up_ramp	"true": drift_thresholds_with_ramp selected for drift calculation. "false", drift_thresholds selected (no consideration of ramp rate)	Mandatory
drift_thresholds_with_ramp	Example of drift thresholds based on current velocity thresholds in m/s, if up_ramp is true (see section 4.4 Metrics calculation)	Based on the value of up_ramp
drift_thresholds	Example of drift thresholds based on current velocity thresholds in m/s, without consideration of ramp rate if up_ramp is false (see section 4.4 Metrics calculation)	

ATTENTION

It is the user’s responsibility to determine the desiccation and drift threshold values that are adapted to the macroinvertebrate taxa and the habitat type of interest in the study. Moreover, the user needs to decide if an up-ramping threshold makes sense for the drift risk calculation.

Optional: advanced parameters for a focus on a specific zone

Name of the variable	Signification	Status
focus_on_zone	"true" if focus set on a specific zone (e.g., gravel banks for eggs desiccation), "false" otherwise (by default)	Mandatory
zone_interest_filename	name of the focus zone shapefile (and other extensions) located in the data>input>Focus_zone folder	Optional
id_polygon_zone_interest	shapefile's id column name of the polygons of interest (e.g., individual gravel bar id, to be able to compare them) /!\ This column needs to be named differently than the shp_id_colname id in the input shapefile. Rename the column if needed	Optional

zone_interest_field	name to give to this focus zone in output files (value of 1 if the patch is part of the focus zone, 0 if not)	Optional

Optional: files paths for plots and clustering (further analysis)

Name of the variable	**Signification**	**Status**
base_output_path	output folder (located in data > output folder) where data to analyse is located in	Optional
final_csv_path	Relative path to the output csv file containing the metrics values	Optional
final_shp_path	Relative path to the output shapefile containing the metrics values	Optional
static_habitat_csv_path	Relative path to the csv file containing the habitat classification of each patch	Optional

Tailoring an analysis

It is encouraged in a first step to run the example case study to get familiar with the toolbox.

In a second step, here are some easy suggestions.
/!\ Save the files that you adapt before running the code for your changes to be considered.

Create a *local.yaml* file (by copy/pasting the *config.yaml*, see video) and try modifying one parameter at a time.

Simple modifications:

- Change the target habitat
- Modify one habitat threshold
- Replace the flow time-series and adapt the timestep accordingly
- Activate or deactivate one metric

More advanced modifications:

- Change desiccation thresholds
- Change drift thresholds
- Run *plots.py* or *clustering.py*
- Change the habitat classification (adapt the *local.yaml* and function attribute_habitat_current_based in habitat_ classification.py accordingly)

Modifications to change the focus of the analysis: focusing on a zone

- Set focus_on_zone to true
- Add a focus zone shapefile and fill in the necessary variables in the *local.yaml* file (zone_interest_filename, id_polygon_zone_interest, zone_interest_field)

- Adapt the habitat classification method to the scope of the analysis

Finally, the goal is to run your own tailored analysis matching your needs. Here are the steps to follow:

- Add your input files in the input folders (such as done in the example case)
- Create a *local.yaml* (by copy/pasting the *config.yaml*, see relevant video), adapt it to your needs, and make sure to save it (Ctrl+S)
- If the habitat classification method has changed, adapt the relevant function in habitat_classification.py accordingly. Make sure to save your changes (Ctrl+S)
- Run the main.py
- Adapt the plots.py to your new habitats (names and colours). Make sure to save your changes (Ctrl+S) before running it.
- Run the clustering.py if you want
- Import the output shapefile to a GIS software (use the default layouts if you use QGIS) and explore the results

4. Full workflow explanation

This chapter explains what happens inside *main.py*, from input data to final metrics.

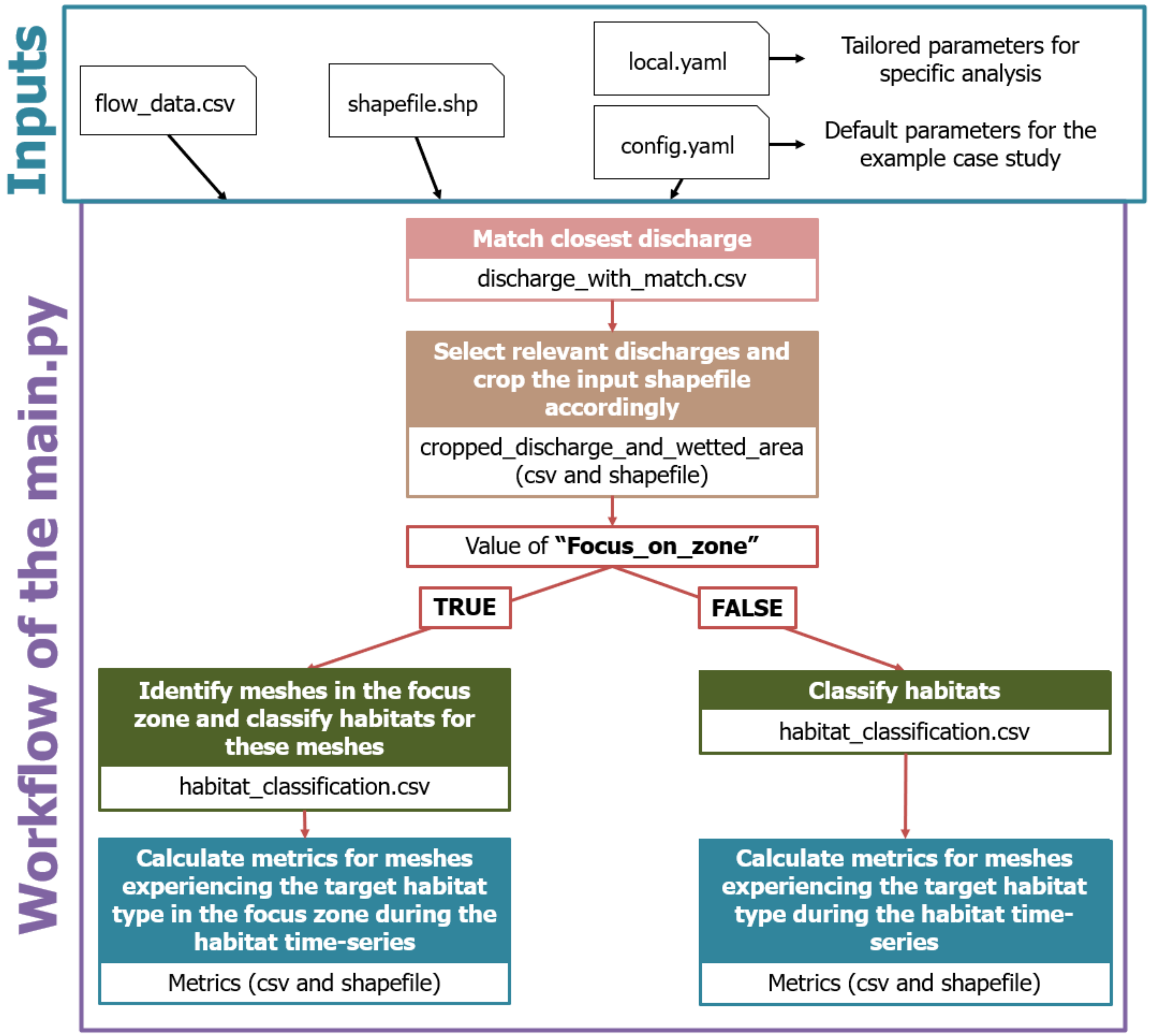


Figure 7 Full workflow of the main.py file

4.1 Match flow time-series to the closest simulated discharges

The toolbox associates the discharges from the input flow time-series to the closest one for which current velocity and depth data is known (data from the input shapefile, see section 3.1 Flow time-series input data). The column "Corresponding_known_discharge" will then be added as seen on Figure 8.

The toolbox needs known water depth and current velocity data for the discharges of interest, given by the shapefile input data. **It is not interpolating data for other discharges**. If more precision is needed, or if more extreme discharges need to be considered, then it means that the input data needs to be adapted by the user.

Datetime	Discharge	Corresponding_known_discharge
1/6/2025 0:00	1.597203229	2
1/6/2025 0:10	2.260468285	2
1/6/2025 0:20	11.27516493	10.6
1/6/2025 0:30	13.44403064	13.4
1/6/2025 0:40	11.49950344	12
1/6/2025 0:50	9.056749663	9

Figure 8 Example of processed flow time-series where each discharge is associated to the closest known one

4.2 Select relevant discharges and crop the input shapefile

From these "Corresponding_known_discharge", the toolbox gets the list of the discharges to keep for the scope of the study. To save computational time, input files get cropped accordingly to keep:

- Only data corresponding to discharge in the scope of the study
- Only patches that are wet under the maximum considered discharge

4.3 Classify habitats

It is the user's responsibility to define how to classify habitat types to match the scope of the analysis. Figure 9 presents the water depth and current velocity-based classification used in the example case. Similar approaches can be used with a varying number of habitat types and different threshold values, **all to be defined and adapted in the configuration file (*local.yaml*) and the *habitat_classification.py* file**.

Color	Current Velocity Class
	Dry
	Stagnant to slow current $v_{40} < 5$ cm/s
	Slow current $5 \leq v_{40} < 25$ cm/s
	Slow to fast current $25 \leq v_{40} < 75$ cm/s
	Fast to very fast current $75 \leq v_{40} < 150$ cm/s
	Very fast current $150 \leq v_{40} < 250$ cm/s
	Violent current $v_{40} \geq 250$ cm/s

Figure 9 Example of macroinvertebrate habitat classification at community-level (adapted from Tonolla 2023)

Another example of habitat classification focusing on a specific zone (with the boolean *focus_on_zone* set to "true") is provided in the function *attribute_habitat_types_zone_only* in the *habitat_classification.py* script. This example targets egg deposition on gravel banks. Habitat type -1 indicates that the patch lies outside the zone of interest (being the gravel banks in our case), while the other habitat types are illustrated in Figure 10.

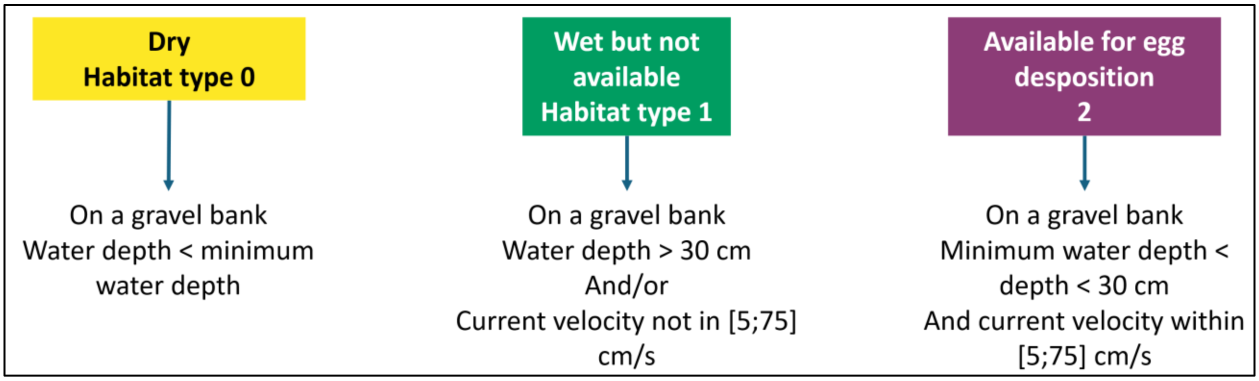


Figure 10 Example of habitat classification focusing on egg desiccation on gravel banks

ATTENTION

When adapting the code in the repository (e.g., *the habitat_classification.py*), make sure to **save your changes before running the code** for your changed to be considered (simple Crtl+S in the file).

As previously mentioned, also make sure to save your changed files to a safe location before pulling the latest version of the Git repository.

4.4 Metrics calculation

Scope of the metrics calculation

The simulated domain includes all patches that are wet under the maximum discharge of the flow time-series. The habitat probabilities are calculated for all these wetted patches.

The metrics are calculated only for patches that experience the habitat type in focus during the habitat time-series. The user can decide which metric(s) to calculate in the configuration file.

Description of the metrics

The macroinvertebrate patch-scale habitat metrics on Table 5 Overview of the four patch-scale metrics used to study macroinvertebrate habitat dynamics, including definitions and calculation are derived from the habitat time-series to quantify complementary ecological facets of habitat dynamics experienced by macroinvertebrates. Habitat probability (M1) and shifts (M2) describe the availability and temporal persistence of suitable conditions and are therefore generally related to habitat quality, whereas drift (M3) and desiccation (M4) metrics represent exposure to harmful hydraulic events with associated ecological risk.

Table 5 Overview of the four patch-scale metrics used to study macroinvertebrate habitat dynamics, including definitions and calculation

Metric	How ?	What ?	Why ?
1) Habitat probability	**Likelihood of different habitat conditions within a patch through time** (Bätz et al. 2025). The value is continuous. Value $= \frac{\sum duration\ of\ occurence\ of\ a\ specific\ \mathrm{habitat}\ type\ over\ the\ time-series}{total\ time\ of\ the\ flow\ time-series}$	0 → 1 Min → Max	- Time-integrated **habitat availability** - Helps to **spot most probable habitat types**
2) Habitat shifts	**Daily number of times (i.e., frequency) that habitat conditions change to another suitable habitat type in a patch** (Bätz et al. 2025). The value is continuous. Value = $\frac{\sum number\ of\ relevant\ \mathrm{shifts}\ during\ \mathrm{the}\ considered\ time-series}{number\ of\ days\ of\ the\ flow\ time-series}$	0 → Daily time intervals Min → Max	- Highlights **habitat stability and persistence**
3) Drift risk	**Risk of macroinvertebrate drift based on current velocity conditions and water level rise experienced within each patch over time.** The value is discrete. Up-ramping can be included in the analysis or not Value = 90^{th} percentile risk value reached over the habitat time-series	1 → 4 Min → Max	- Identifies exposure to critical **passive drift**
4) Desiccation risk	**Risk of eggs desiccation linked to hydropeaking-induced water level variations.** The value is discrete, and calculated from start or after a first occurrence Value = maximum risk value reached over the habitat time-series***Error! Reference source not found.***	1 → 4 Min → Max	- Identifies exposure to critical desiccation **duration**

Drift and desiccation risks are based on threshold values, that depend not only on the considered taxa, but also on the available data and the habitat type in focus.

Table 6 and Table 7 are examples of threshold values used in example case.

Table 6 focuses on the drift risk for two habitat types: Slow (habitat type 2) and Slow to fast (habitat type 3) current velocity. “Slow” habitat type considers the notion of up-ramping rate (up_ramp set to “true” in the configuration file, and a value entered in drift_ramp_threshold). The analysis of the habitat type “Slow to fast” has no up_ramp. The parameters set in the configuration file allow to pick for the right drift calculation function in the metric calculation scripts. It is up to the user to decide weather the up-ramping rate needs to be considered in the analysis or not, given the available data and the scope of the study.

Table 7 presents the example of egg desiccation risk threshold values implemented in the toolbox.

Table 6 Macroinvertebrate drift risk thresholds and corresponding metric values for two considered habitat types

Metric Value	Drift risk class	Habitat type		
		Slow [a]		Slow to fast [b]
		Current velocity threshold	Water level rise	Current velocity threshold
1	Low	$v_{40} \leq 0.4$ m/s	Not relevant	$v_{40} \leq 1.2$ m/s
2	Moderate	$0.4 < v_{40} < 0.7$ m/s	Not relevant	$1.2 < v_{40} < 2.1$ m/s
		$0.7 \leq v_{40} \leq 1$ m/s	< 1cm/min	
3	High	$0.7 \leq v_{40} \leq 1$ m/s	≥ 1cm/min	$2.1 \leq v_{40} \leq 3$ m/s
4	Very high	$v_{40} > 1$ m/s	No data	$v_{40} > 3$ m/s

[a]: thresholds taken form the flume experiments in Tonolla et al. (2019) and incorporating findings from Friese et al. (2025).
[b]: thresholds mathematically extrapolated from the "slow" current habitat, by proportionally adjusting the upper limit of the typical current velocity range. No water level rise rate included to avoid introducing additional assumptions.

Table 7 Macroinvertebrate desiccation risk thresholds and metric values based on Kennedy et al. (2016) and Jaulin (2025)

Metric value	Desiccation risk class	Max duration dry
1	Low	≤ 1 hour
2	Moderate	1 - 5 hours
3	High	5 - 9 hours
4	Very high	> 9 hours

The *start_at_first_occurrence* boolean can change the values of the desiccation risk: if set to "true", desiccation risk will only be calculated after the first occurrence of the target habitat type. If set to false, the calculation starts at the beginning of the flow time-series. For example, for a study focusing on egg desiccation on gravel banks, setting the *start_at_first_occurrence* to true will make the desiccation risk calculation start after the first occurrence of the target habitat type that is "suitable for egg deposition".

Advanced metrics when focusing on a certain zone

In the advanced case when *focus_on_zone* is true, additional metrics such as median and quartile values are being calculated for the habitat type in focus and the dry conditions to have a deeper understanding of the habitat probability distributions (see the Table 11 in section 5.1 Outputs from the main file execution, and the related part of section 5.2 Outputs from the plot execution).

4.5 Estimation of the computational time

The computational time can drastically vary depending on the spatial and temporal coverage of the analysis (see estimations on Figure 8).

Table 8 Estimation of computational times based on the number of patches and of timesteps included in the analysis

Number of patches (total)	Number of patches for metrics calculation	Number of timesteps	Estimated computational time
30 446	25 769	1008	~ 5 minutes
365 000	27 350	1008	~ 50 minutes
664 000	76 060	4393	~ 4 hours

4.6 Additional functions

Table 9 Association between the code to run and the desired outputs

Output to get	Python file to run
Metrics results	*main.py*
Plots	*plots.py*
Clustering	*clustering.py*

Before running the additional scripts, paths to the files to study (*base_output_path, final_csv_path, final_shp_path, static_habitat_csv_path*) need to be entered in the configuration file (see section 3.2), which then needs to **be saved** before running the scripts.

The metrics in the scope of the plot and the clustering functions are taken from the configuration file. This means you must ensure the metrics selected match the ones that are in the files to analyze. E.g., if the boolean *desiccation_risk* is set to true in the configuration file, it also needs to be in the file to analyze because the functions will be searching for it.

Plots

Outputs location: in data > outputs > base_output_path > Plots

Plots focus on two geographical extents:

- The entire simulated domain, for which habitat probabilities and most probable habitat types are assessed. This helps understanding the global picture and understand why certain patches do not experience the focus habitat type during the habitat time-series.
- The domain for which metrics are calculated, including only patches that experience the focus habitat type at least once during the habitat time-series.

If more targeted plots are needed, the user is free to modify the functions or add new ones (/!\ as previously mentioned, pulling the latest version of the repository from GitHub will overwrite everything so make sure to save your changes in a safe folder).

Examples of outputs are in section 5.2 Outputs from the plot execution.

Clustering

<u>Outputs location:</u> in data > outputs > base_output_path > Clustering

This script conducts a k-means clustering to synthesize the combined information contained in the four metrics. The purpose is to group patches with similar multimeric profiles, thereby summarizing complex combinations of macroinvertebrate habitat quality and risk into a reduced number of ecologically interpretable groups. Because each metric represents a distinct dimension of macroinvertebrate habitat dynamics, clustering enables the identification of patterns that are not apparent when examining each metric separately. The optimal number of clusters is determined using the silhouette score, with maximum numbers of clusters set to 9 (can be adapted in the code if needed).

The functions provide details on the cluster statistics, on the Principal Component Analysis (PCA) loadings, and adds the cluster number of each patches in the csv and shapefile metrics documents. More information on the output files is available on section 5.3).

5. Outputs interpretation

Outputs location: data > output > base_output_path folder. Relevant subfolders separate the outputs.

5.1 Outputs from the main file execution

The *main.py* file execution provides the outputs presented in Table 10.

Table 10 Outputs of the main.py file

Output name	Output location	Explanation
discharge_with_match.csv	Discharge_with_match	Displays each discharge of the flow time-series with its corresponding known simulated discharge.
cropped_discharge_and_wetted_area (csv and shapefile)	Cropped_shapefiles	Input shapefile cropped to the considered discharges and the associated spatial area (patches that are wet under the maximum discharge considered). This allows to reduce the zone of interest before processing the metrics to reduce the computational time.
habitat_classification.csv	Habitat_classification	Static picture of habitat classification for each patch and each considered discharge before time integration. This file is used for the plots to represent habitat availabilities at each discharge.
metrics (csv and shapefile)	Metric_files	Files having the final metrics results for each patch experiencing the habitat target during the habitat time-series.

The shapefile column names are limited to 10 characters. Table 11 provides more details on shapefile column names that had to be shrunk.

In the following table, the "h" stands for habitat, and will always be followed by a number in the output metrics.csv file. As previously mentioned, the metrics are only calculated for patches experiencing the target habitat type at least once during the habitat time-series.

Table 11 Explanation of the different columns that can be found in the metric outputs (csv and shapefile)

Id	Id of the patch (identical as input shapefile)
x	X coordinate of the patch (identical as input shapefile)
y	Y coordinate of the patch (identical as input shapefile)
area	Area of the patch (identical as input shapefile)

d_discharge	Depth value for each discharge in scope of the current analysis (identical as input shapefile)
v_discharge	Current velocity value for each discharge in scope of the current analysis (identical as input shapefile)
focus_zone	Only if there is a focus on a certain zone (focus_on_zone is true). 0 if not in the interest zone shapefile, 1 if in the focus zone shapefile
id_focus	Only if there is a focus on a certain zone (focus_on_zone is true). Indicates the id of the polygon of the interest zone shapefile (e.g., the individual id of the gravel bank)
Habitat probability	
dur_h	Duration of each habitat (in number of timesteps) during the habitat time-series The amount of these columns depends on the habitat classification: - From -1 to 3 for the egg desiccation habitat classification example - From 0 to 6 for the example habitat classification based on water depth and current velocity
prob_h	The associated habitat probabilities = duration habitat type/total duration (in number of timesteps)
mostProb	Most probable habitat type per patch overt the habitat time-series, corresponding to the habitat type having the highest probability. This is calculated for each patch of the simulated domain, not only the ones experiencing the target habitat type (see Figure 11)
h_first	Timestep number of first occurrence of the target habitat type. e.g., h2_first: timestep number of the first occurrence of the habitat type 2 for this patch over the habitat time-series
Statistics on the target habitat type – only for advanced studies focusing on a certain zone	
h_nb_seq	Number of separated sequences of target habitat type conditions in the patch over the habitat time-series
h_dur_max	Maximum consecutive duration (in hours) of target habitat type conditions in the patch over the habitat time-series
h_dur_med	Median of the consecutive duration (in hours) of target habitat type conditions in the patch over the habitat time-series
h_dur_q1	1st quartile of the distribution of the consecutive duration of target habitat type conditions in the patch over the habitat time-series (in hours)
h_dur_q3	3rd quartile of the distribution of the consecutive duration of target habitat type conditions in the patch over the habitat time-series (in hours)
Habitat shifts	
h_sh_all	Total daily shifts experienced by this patch over the habitat time-series

h_sh_suit	If not focusing on a zone: total daily shifts to the suitable habitat types (entered in the configuration file) over the habitat time-series. If focusing on a zone: total daily shifts to the target habitat over the habitat time-series
h_sh_dry	Total daily shifts to dry conditions over the entire habitat time-series
Desiccation	
h_dryMax	Maximal duration of consecutive dry conditions (in number of timesteps). The start of the calculation depends on if start_at_first_occurence is "true" or "false": calculation will start at the beginning of the habitat time-series if "false", and only after the first occurrence of the target habitat type if "true"
h_desicR	Associated desiccation risk
Desiccation additional statistics – only for advanced studies focusing on a certain zone	
h_dry_seq	Number of separated sequences of dry conditions in the patch over the habitat time-series
h_dry_max	Maximum consecutive duration of dry habitat type conditions in the patch over the habitat time-series (in hours)
h_dry_med	Median of the consecutive duration of dry habitat type conditions in the patch over the habitat time-series (in hours)
h_dry_q1	1st quartile of the distribution of the consecutive duration of dry habitat type conditions in the patch over the habitat time-series (in hours)
h_dry_q3	3rd quartile of the distribution of the consecutive duration of target habitat type conditions in the patch over the habitat time-series (in hours)
h_dryMedR	Risk value associated to the median of the consecutive duration of dry habitat type conditions in the patch over the habitat time-series
h_dry_q1R	Risk value associated to the 1st quartile of the distribution of the consecutive duration of dry habitat type conditions in the patch over the habitat time-series
h_dry_q3R	Risk value associated to the 3rd quartile of the distribution of the consecutive duration of target habitat type conditions in the patch over the habitat time-series
Drift	
h_dd_driftRisk	Drift duration of each drift risk category (in number of timesteps)
h_driftP	Associated drift risk percentile
h_driftM	Associated maximal drift risk value

5.2 Outputs from the plot execution

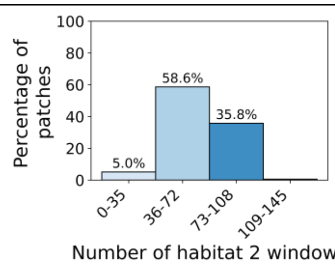
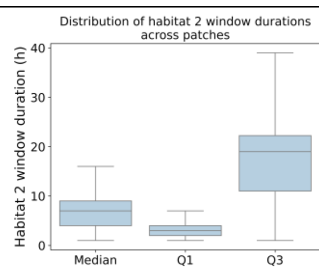
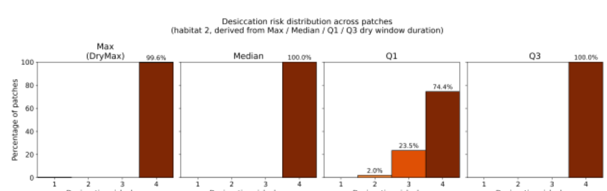
Default plot outputs are presented and explained for the example case study in the following section 5.4 Example case study: output presentation.

Output location: data > output > base_output_path > Plots

The description of the plot function is available in section 4.6 Plots. The user can always modify the function to tailor plots to specific needs, add or delete some. As previously mentioned, the updated *plot.py* needs to be saved in a separate file not to be overwritten when pulling the latest version of the GitHub.
The outputs depend on the values of the parameters in the *local.yaml* configuration file. For the case when patches have different sizes, some plots normalize by area.

For advanced analysis such as the ones focusing on zones (Boolean *focus_on_zone* is true), additional plots can be gone as presented on the below. To compare the different gravel banks, it means that all patches have an id corresponding to which gravel bank they belong to.

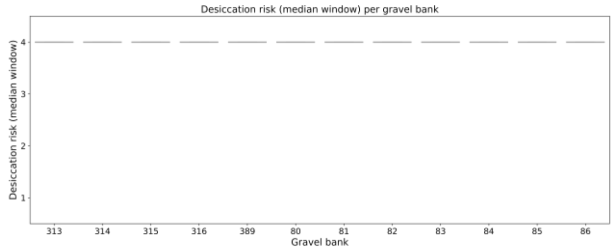
Table 12 Advanced plots done by the file plot.py when focusing on a specific zone

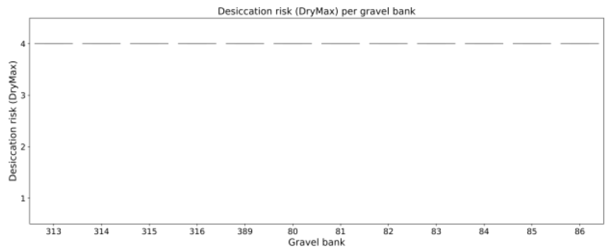
Target habitat: statistics to add up to the	
Number of target habitat sequence individual windows	
Distribution of target habitat sequences durations (in hours)	
Dry durations and desiccation risk	
Desiccation risk Distribution of the risk values across patches	

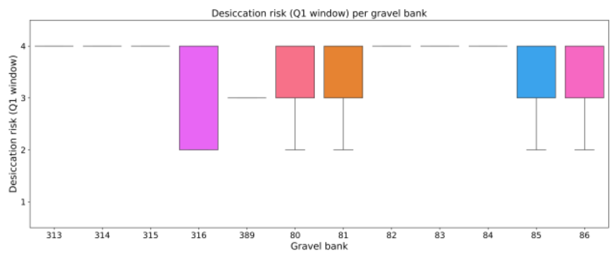
<table>
<tr>
<td>Median desiccation risk per gravel bank

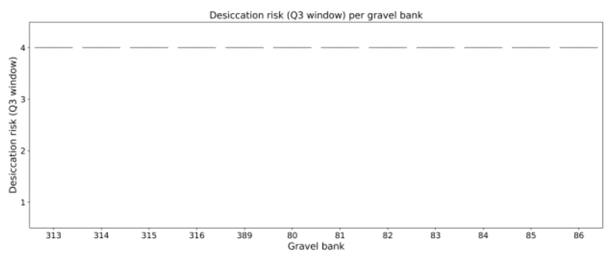
Desiccation risk per gravel bank

Q1 of the desiccation risk per gravel bank

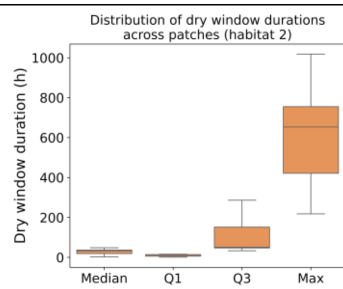
Q3 of the desiccation risk per gravel bank</td>
<td>


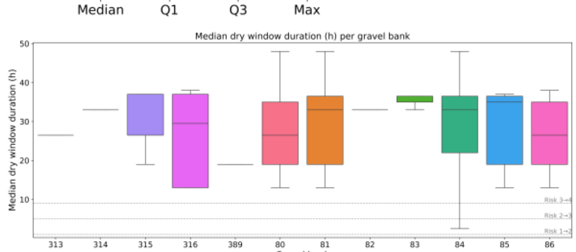




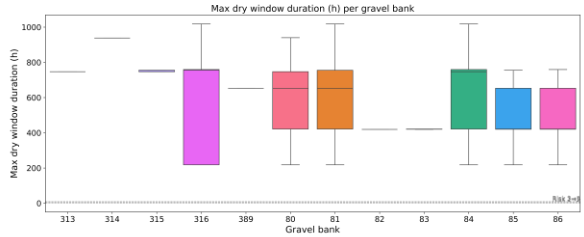


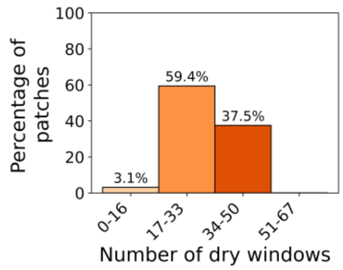
</td>
</tr>
<tr>
<td>Associated dry window durations

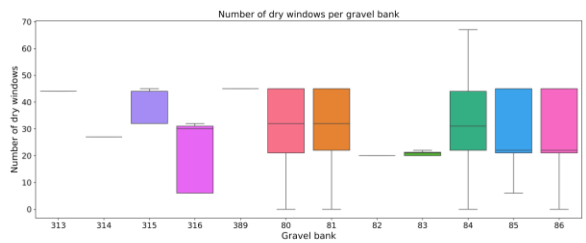
Median dry window duration (in hours) per gravel bank</td>
<td>




</td>
</tr>
</table>

<table>
<tr>
<td>Maximum dry window duration (in hours) per gravel bank</td>
<td>
</td>
</tr>
<tr>
<td>Number of dry windows

Number of dry windows per gravel bank</td>
<td>


</td>
</tr>
</table>

5.3 Outputs of the clustering execution

Output location: data > output > base_output_path > Clustering

Description of the plot function available in section Additional functions 4.6 Clustering.

The outputs depend on the values of the parameters in the *local.yaml* configuration file.

Outputs are presented for the example case study in the following section 5.4 Example case study: output presentation.

5.4 Example case study: output presentation

Metrics outputs and visual representation

A QGIS project representing the metrics outputs is available in the data > output > Example_output > QGIS_project foler. An explanatory video on the topic is also available.

To ease the visualization and the interpretation, an example template of a map is also available in the same folder (example_case_maps_template.qpt). Default styles for metrics representation on QGIS are also available.

For ArcGis user, the example case study is also available in the same folder under HaDy_MZB_Arcgis.aprx.

Example case study: Metrics outputs and visual representation

As presented on Figure 11, the most probable habitat type over the habitat time-series are represented in a layer, such as the four metrics for the default parameters from the *conifg.yaml* file.

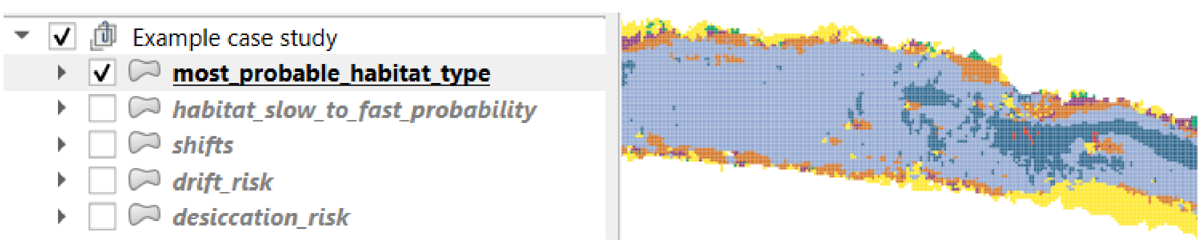


Figure 11 QGIS layers present for the example case study

Plot outputs

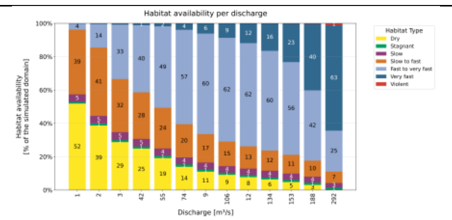
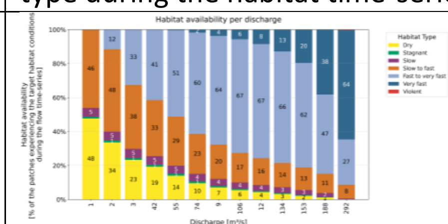
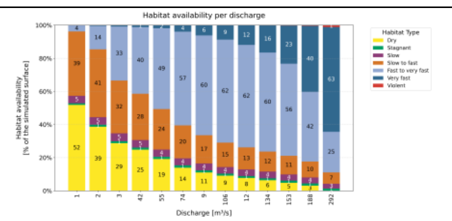
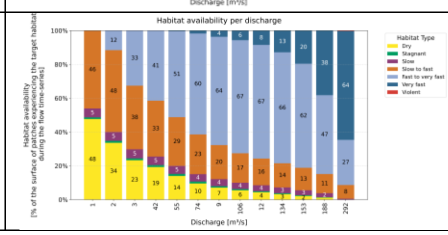
Example case study: Example case study plots

Table 13 below presents examples of plots done by the file *plot.py* for the example case study.

Table 13 Plots done by the file plot.py for the example case study

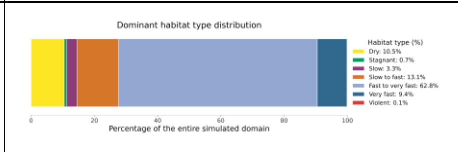
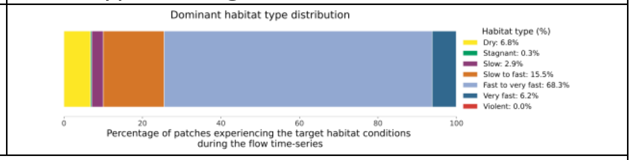
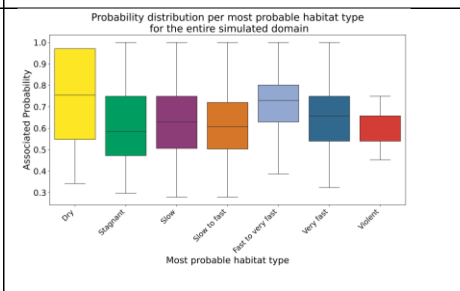
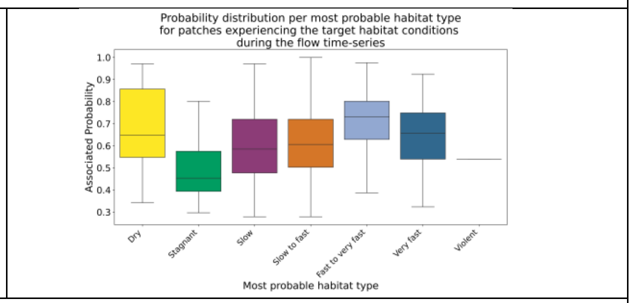
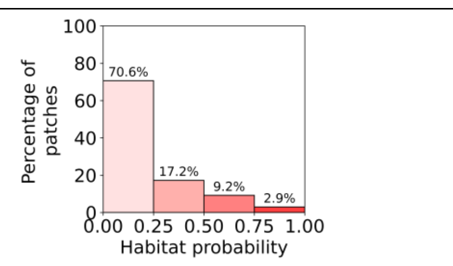
<table>
<tr><td colspan="3">Static representation of habitats distribution per discharge
The different considered discharges are in the x axis. For each one of them, the static picture of the habitat conditions is presented.

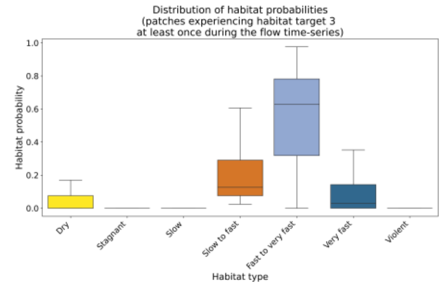
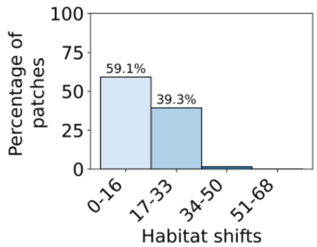
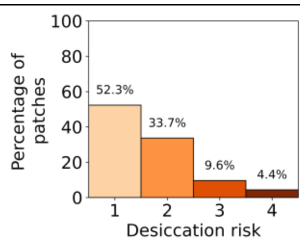
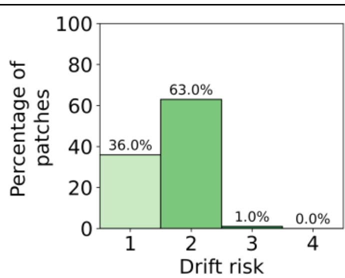
Lines: since patches do not necessarily have the same area, the figures are available per number of patches, but also per area that it represents.

Columns: the habitat distribution is available for two spatial coverages: the entire simulated domain (covering all patches that are wet under the maximum considered discharge), but also only the patches that experience the target habitat during the habitat time series.</td></tr>
<tr><td></td><td>For the entire simulated domain</td><td>For patches experiencing the target habitat type during the habitat time-series</td></tr>
<tr><td>Per number of patches</td><td>
</td><td>
</td></tr>
<tr><td>Per area</td><td>
</td><td>
</td></tr>
<tr><td colspan="3">Most probable habitat type over the habitat time-series</td></tr>
<tr><td colspan="3">For each patch, the most probable habitat type is the habitat type having the highest probability over the habitat time series.
The probability of each habitat type: time the patch spends in this habitat type over the flow time-series / total length of the flow time-series.

The figures below represent what is the general distribution of the most probable habitat types over the considered flow time-series. This is an integration in space and in time.

Columns: the same as before, meaning the left represents the entire simulated domain (covering all patches that are wet under the maximum considered discharge), and the right only the patches that experience the target habitat during the habitat time series.</td></tr>
</table>

<table>
<tr><td></td><td>For the entire simulated domain</td><td>For patches experiencing the target habitat type during the habitat time-series</td></tr>
<tr><td rowspan="2">Most probable habitat type distribution (per number of patches)</td><td>
</td><td>
</td></tr>
<tr><td colspan="2">This helps understanding what habitat type is most likely to be found over the habitat time-series. This example shows the habitat type “fast to very fast” is highly represented: it is the most probable habitat type of 62.8% of the patches of the simulated zone</td></tr>
<tr><td rowspan="2">Associated probability distributions of each most probable habitat type</td><td>
</td><td>
</td></tr>
<tr><td colspan="2">This provides information on the associated probabilities: for each patch, the most probable habitat type is the one having the highest probability. But what is that associated probability? Is it 0.2? 0.5? This helps understanding what is happening over time. In this example, the probability of the “fast to very fast” dominated patches is having the median probability value of around 0.75, which is quite high. This is consistent with the flow time-series conditions impacted by hydropeaking. This means that, over time, a large majority of the riverbed will regularly experience “fast to very fast” conditions.</td></tr>
<tr><td colspan="3">Metrics representations
Only for patches experiencing the target habitat type during the habitat time-series
See section 4.4 for more information on the metrics calculation</td></tr>
<tr><td></td><td>For patches experiencing the target habitat type during the habitat time-series</td><td>Explanation</td></tr>
<tr><td>Habitat probability of the target habitat type</td><td>
</td><td>This represents the probability values of the target habitat type.
In our example case, the target habitat type is “slow to fast”. The patches experiencing it only spend little time in it (87.8% of them spend less than 50% of their time in this target habitat type conditions).</td></tr>
</table>

<table>
<tr><td>Habitat probabilities distributions</td><td>
</td><td>By showing the distributions of each habitat type probabilities of these patches, it provides additional information. As expected from previous figures, they spend most of their time in the “fast to very fast” habitat conditions.</td></tr>
<tr><td>Daily shifts to defined habitats</td><td>
</td><td>This represents the total daily shifts to the suitable habitat types (entered in the configuration file) over the habitat time-series. In our case, we considered stagnant, slow, slow to fast, fast to very fast, and very fast as suitable. This means that a shift is calculated when there is a change from an unsuitable habitat type (dry or violent) to a suitable habitat type.</td></tr>
<tr><td>Desiccation risk</td><td>
</td><td>See section 4.4 for more information on how the risk is calculated. This represents the percentage of the patches that experience each desiccation risk level.</td></tr>
<tr><td>Drift risk</td><td>
</td><td>See section 4.4 for more information on how the risk is calculated. This represents the percentage of the patches that experience each drift risk level.</td></tr>
</table>

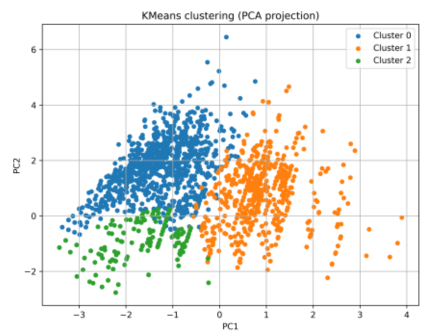
Clustering outputs

Example case study: Example case study clustering

below presents examples of plots done by the file *clustering.py* for the example case study.

Table 14 Clustering outputs done by the file clustering.py for the example case study

Output	Description
Hab_silhouette.png	Plot showing the evolution of the silhouette score with the number of clusters. The maximum value determines the optimal number of clusters.

	
Hab_PCA_clusters.png	Graphically represents the position of the clusters on the PC1/PC2 axis 
Hab_PCA_loadings.csv	Provides the value of PC1 and PC2 loadings for each metric
metrics_with_clusters (csv and shapefile)	Same outputs as the metric results ones, having additional columns: - Cluster_hab: indicating the cluster number of each patch - PC1_hab and PC2_hab: principal component loadings of the patch

6. Limitation and assumptions

Input shapefile: it is assumed that the user has identified key study discharges and conducted a hydrodynamic simulation to get the required input shapefile.

Habitat classification and metrics threshold values: the toolbox only provides example threshold values that need to be tailored to the user needs and taxa of interest.

GIS analysis: the visual analysis of the output shapefile is to be done on a GIS tool by the user.

7. Trouble shooting

Here is a list of recommendations to get familiar with the toolbox and avoid troubles.

7.1 Start small, in space and in time

Start with small spatial zones and short flow time-series: this will not take long to compute, and you will still be able to validate the results with common sense before it gets too complex.

To help you crop the original shapefile to a smaller zone to get started, the QGIS procedure is explained in the section 9.2 Crop a .shp to a specific shape in QGIS.

7.2 Save your scripts and analysis files

As previously mentioned, when pulling the latest version of the GitHub repository, your current data and scripts will be overwritten. Make sure to save your files in a separate folder not to lose anything.

7.3 Follow the instructions carefully

When it is said that the column names format matters, it is because the code extracts an information out of them. For instance, if your input shapefile column names are not depth_18_3 for the discharge 18.3 m^3/s (as requested), but entered as depth_18.3, the code will crash, and you will have no outputs. You can simply modify your column names with your usual GIS tool such as QGIS.

Moreover, if you do not create the *local.yaml* file, the toolbox will run with the default parameters (corresponding to the example case study). Make sure to respect the necessary steps to tailor and conduct specific analysis.

8. Glossary

Patch: is a spatially fixed unit within a river reach, typically corresponding to areas on the order of 1 to 100 m² (micro- to meso-scale), with the appropriate size depending on the mobility, species, and life stage considered. While the location of a patch remains constant, the habitat conditions within it may vary over time. In this toolbox, the patches usually match the meshes of the hydrodynamic simulation for which depth and current velocity data is available.

Habitat: refers to a set of environmental conditions characterized by dominant physical and chemical properties (e.g., current velocity, water depth, substrate). These conditions can be described at different spatial scales, including the micro- to meso-scale (1–100 m²) within a river reach. Different species and life stages rely on specific habitat conditions to fulfil essential functions such as feeding, reproduction, resting, or refuge. The spatial distribution of habitats within a river reach can change over time.

Habitat shift: occurs when environmental conditions change in space and/or time, leading to a transition from one habitat type to another. In this toolbox, habitat shifts are assessed at the patch scale, capturing temporal changes in habitat conditions within a fixed spatial unit.

Flow time-series: describes the temporal evolution of river discharge (m^3/s), represented at a defined timestep (e.g., minutes or hours).

Habitat time-series: represents the temporal sequence of habitat conditions within each patch, obtained by combining spatial habitat classifications with the flow time-series.

Target habitat: The habitat type that is the primary focus of the analysis. Habitats are defined according to a chosen classification scheme, where each type is assigned a numerical code for processing within the toolbox. Metrics are computed specifically for patches that experience the target habitat at least once during the habitat time-series.

Suitable habitats: Habitat types that are considered favourable for the macroinvertebrate taxa under study, based on the criteria defined in the selected habitat classification scheme.

Drift percentile (drift risk): A patch-scale metric representing exposure to hydraulically induced drift. It is defined as the 90th percentile of all drift-risk values computed from the habitat time-series for a given patch. This percentile-based approach captures frequent exposure to elevated drift conditions while reducing the influence of isolated extreme events.

Desiccation risk: A patch-scale metric representing exposure to drying conditions over time. It is derived from the habitat time-series and reflects the occurrence and persistence of conditions where habitat becomes unsuitable due to insufficient water depth, indicating ecological stress for macroinvertebrates.

9. GQIS tips and tricks

9.1 Analysing a shapefile field on QGIS

To focus on a specific shapefile field, the Figure 12 Example of how to process the output shapefiles on QGIS is an example of how to do it on QGIS. You can also use the default symbology done for the different metrics available in outputs > example_output > GIS_project (all the .qml symbology files).

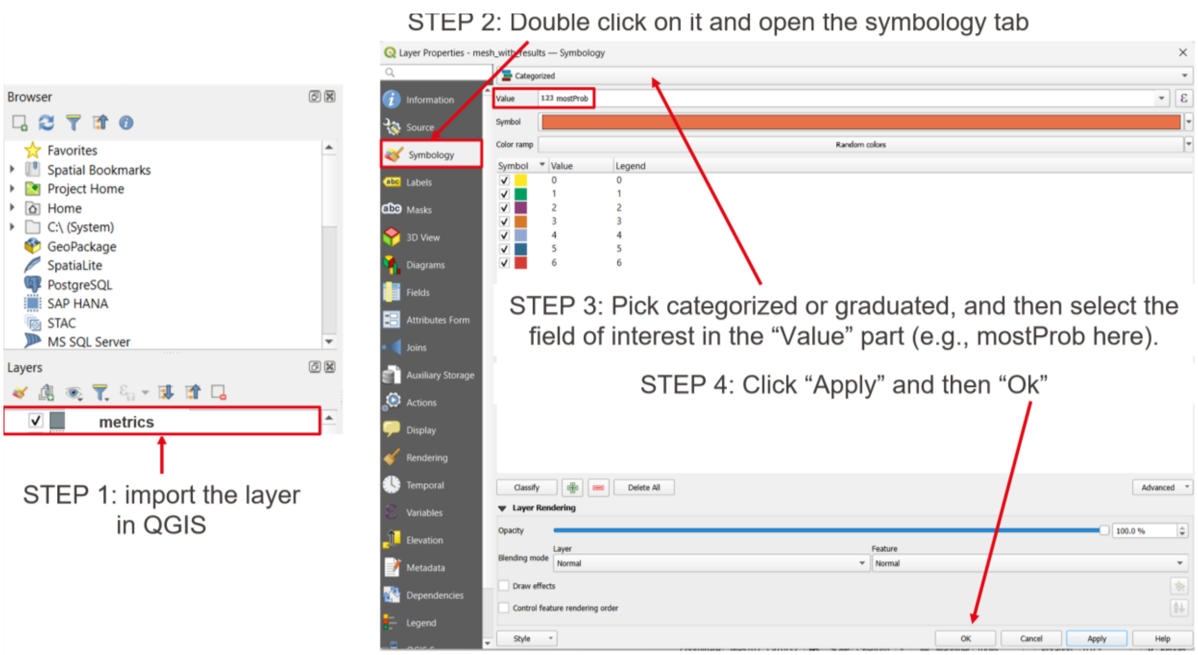


Figure 12 Example of how to process the output shapefiles on QGIS

9.2 Crop a .shp to a specific shape in QGIS

This gives a reduced .shp file containing the data for the area of interest.

Step 1. Prepare your layers

- You need two layers:
    1. The .shp layer to crop (the one you want to limit to a certain area).
    2. The mask layer (the polygon defining the area you want to crop to).

Step 2. Use the "Clip" tool

1. Open the Processing Toolbox (from the menu: Processing → Toolbox).
2. Search for "Clip".
3. Choose Vector overlay → Clip.
4. In the dialog:
    - Input layer: the shapefile you want to crop.
    - Overlay layer: the polygon mask.
    - Save to file: choose a path for the new cropped shapefile.
5. Click Run.

- The resulting shapefile will only contain features inside your mask polygon.

Step 3. About features with missing data

- If parts of the shapefile inside your mask have null or missing attribute values, the clip will still work.
- QGIS doesn’t require all attributes to be filled; missing data won’t stop the crop.
- However, if you need to do further analysis (e.g., statistics or joins), you may need to handle nulls.

Step 4. Optional: use "Extract by location"

- If you only want features fully within your mask polygon, you can use Vector → Geoprocessing Tools → Extract by location, choosing “within” instead of “intersect”.
- This avoids partially overlapping features if you only want complete ones.

9.3 Joining layers on QGIS

To be done when two input shapefiles need to be joined into a single one (e.g., one is a point .shp having depth and velocities values for each patch, the other one is a polygon .shp just containing the shape of the patches). Both have the same center of the polygons, to be able to join the attributes by location.

The desired output is one polygon .shp file having data for every patch.

Necessity: having a common id between the two input shapefiles to join based on it.

How to do: Perform the “create spatial index” for both layers: layer > source > geometry > create spatial index.

Then in the processing toolbox, process the function: **“join attributes by location”**.

10. Appendix

10.1 Deep description of files and functions

Main.py

The **main.py** file collects functions from other python files and variables entered by the user in the configuration file and executes the code.

Description of the workflow (in chronological order)	Name of the function	Where the function is located	Name of the output
Getting the known discharge values from the column names of the input polygon shapefile	get_discharge_values	*support_functions.py*	discharge_values
Match the discharge of the flow time-series to the closest known one in discharge_values	match_closest_discharge	*flow_time_series.py*	discharge_with_match.csv
Select relevant discharges for the analysis: the list of corresponding known discharges	get_study_discharges	*flow_time_series.py*	relevant_discharges
Crop the shapefile: to the corresponding relevant discharges; and keeping only the wetted mask (patches wet for the max considered discharge)	prepare_wetted_shapefile_for_relevant_discharges	*support_functions.py*	cropped_discharge_and_wetted_area.shp
Export the cropped shapefile to a csv	prepare_csv	*habitat_classification.py*	cropped_discharge_and_wetted_area.csv
If focus_on_zone is true			
Attribute habitat only for patches in the focus zone (otherwise -1 by default). This habitat classification function needs to be adapted by the user	attribute_habitat_types_zone_only	*habitat_attribution.py*	habitat_classification.csv
Metrics calculation	process_mesh_data_focus_on_zone	*metrics_calculation_focus_on_zone.py*	metrics.csv

Save the metrics to a shapefile	join_mesh_with_CSV_data	*support_functions.py*	metrics.shp
If focus_on_zone is false			
Attribute habitats based on the example habitat classification	attribute_habitat_current_based	*habitat_classification.py*	habitat_classification.csv
Metrics calculation	process_mesh_data	*metrics_calculation.py*	metrics.csv
Save the metrics to a shapefile	join_mesh_with_CSV_data	*support_functions.py*	metrics.shp

Plot.py

The ***plot.py*** contains the function for the different plots. The user can always add and/or modify these functions (more information in section 4.6 Additional functions and section 5.2).

Description of the workflow (in chronological order)	Name of the function	Name of the output
ALL PATCHES IN SIMULATED DOMAIN Plot most probable habitat type and related intensity	*plot_most_prob_intensity*	hist_most_prob_and_associated_intensity_simulated_zone.png
TARGET HABITAT Plot most probable habitat type and related intensity	*plot_most_prob_intensity_target_hab*	hist_most_prob_and_associated_intensity_target_habitat.png
ALL PATCHES IN SIMULATED DOMAIN Percentage of most probable habitat types	*plot_most_prob_percentages*	most_prob_percentages_simulated_zone.png
TARGET HABITAT Percentage of most probable habitat types	*plot_most_prob_percentages_target_hab*	most_prob_percentages_target_hab.png
ALL PATCHES IN SIMULATED DOMAIN Horizontal stacked bar of dominant habitat types	*plot_most_prob_horizontal*	most_prob_horizontal_simulated_zone.png
TARGET HABITAT Horizontal stacked bar of dominant habitat types	*plot_most_prob_horizontal_target_hab*	most_prob_horizontal_target_hab.png
ALL PATCHES IN SIMULATED DOMAIN IN PERCENTAGE OF PATCHES - Habitat availability per discharge (stacked %)	*plot_habitat_availability_per_discharge*	habitat_availability_per_discharge_simulated_domain.png

TARGET HABITAT IN PERCENTAGE OF PATCHES - Habitat availability per discharge (stacked %)	*plot_habitat_availability_per_discharge_target_hab*	habitat_availability_per_discharge_target_hab.png
FUNCTION: ALL PATCHES IN SIMULATED DOMAIN IN AREA – Habitat availability per discharge (stacked %)	*plot_habitat_availability_per_discharge_area*	habitat_availability_per_discharge_simulated_domain_area.png
TARGET HABITAT IN AREA – Habitat availability per discharge (stacked %)	*plot_habitat_availability_per_discharge_area_target_hab*	habitat_availability_per_discharge_simulated_domain_target_hab_area.png
TARGET HABITAT Habitat probability for each habitat type (Among all patches that experience the target habitat, what is the distribution of time spent in each habitat?)	*plot_probability_distribution_target_hab*	habitat_probabilities_target_hab.png
TARGET HABITAT Plot percentage histograms for selected metrics of a target habitat	*plot_histograms_target_habitat*	histogram_prob_percentage.png histogram_shifts_percentages.png histogram_desicc_percentages.png histogram_drift_percentages.png boxplot_dry_window_duration.png boxplot_targ_window_duration.png
TARGET HABITAT Compare desiccation metrics across gravel banks (id_focus) using Kruskal-Wallis and Dunn post-hoc tests, with boxplots per polygon.	*plot_desiccation_by_polygon*	desicc_max_h_by_polygon.png desicc_median_h_by_polygon.png desicc_risk_by_polygon.png desicc_risk_median_by_polygon.png desicc_risk_q1_by_polygon.png desicc_risk_q3_by_polygon.png

		desicc_window_count_by_polygon.png desiccation_summary_statistics.csv

Clustering.py

The ***clustering.py*** contains the function regarding the clustering presented in section 4.6 Additional functions and section 5.3 . The user can always add and/or modify this file.

Description of the workflow (in chronological order)	Name of the function	Name of the output
Performing the clustering: Finding the best number of clusters, PCA analysis, cluster statistics	*perform_clustering_target_habitat*	metrics_with_clusters.csv PCA_clusters.png PCA_loadings.csv silhouette.png
Join a GeoDataFrame (patch geometry) with a results DataFrame and save to a new shapefile	*join_mesh_with_CSV_data*	metrics_with_clusters.shp

11. References

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